2-17

2–17 At a certain location, wind is blowing steadily at 10 m/s. Determine the mechanical energy of air per unit mass and the power generation potential of a wind turbine with 60-m-diameter blades at that location. Take the air density to be 1.25 kg/m^3.

Given:

- Wind speed: $v = 10$ m/s
- Turbine blade diameter: $d = 60$ m
- Air density: $\rho = 1.25$ kg/m^3

Find:

- Mechanical energy of air per unit mass
- Power generation potential of the wind turbine

Power generation potential refers to the *maximum* amount of power that can be generated by the wind turbine. This occurs if all the kinetic energy of the wind is converted into rotational energy of the turbine blades. Thus, the mechanical energy of the air per unit mass is:

$$E_{\text{mech}} = \frac{1}{2}mv^2 \implies e_{\text{mech}} = \frac{1}{2}v^2$$

$$e_{\text{mech}} = \frac{1}{2}\left[10\,\frac{m}{s}\right]^2 = \left[50\,\frac{m^2}{s^2}\right]\left[\frac{J}{N\cdot m}\right]\left[\frac{N}{kg\cdot m/s^2}\right] = \boxed{50\,\frac{J}{kg}}$$

Power is the *rate* of energy transfer or $\dot{W}$. Thus, multiplying the mechanical energy per unit mass by the mass flow rate $\dot{m}$ gives us energy per unit time, or power:

$$\boxed{\dot{W} = \dot{m}e_{\text{mech}}, \quad \dot{m} = \rho v A}$$

$$\dot{W} = \rho v A \cdot \frac{1}{2}v^2 = \frac{1}{2}\rho A v^3 = \frac{1}{2}\left[1.25\,\frac{kg}{m^3}\right]\left[\frac{\pi}{4}(60\,m)^2\right]\left[10\,\frac{m}{s}\right]^3$$

$$\dot{W} = \left[1767145.86\,\frac{kg\cdot m^3}{s^3}\right]\left[\frac{J}{\frac{kg\cdot m^2}{s^2}}\right] = \boxed{1767\times 10^3\,\frac{J}{s}}$$

We can also find $\dot{W}$ by applying the first law of thermodynamics to the control volume (open system) around the turbine blades:

$$\boxed{\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i\left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e\left[h_e + \frac{1}{2}v_e^2 + gz_e\right]}$$

Where:

- dE/dt is the rate of change of the total energy of the system over time
- $\dot{Q}$ is the rate of heat transfer into the system from the surroundings

- $\dot{W}$ is the rate of work done by the system on its surroundings (power output)
- The summations represent the total energy entering and leaving the system due to mass flow, where h is the specific enthalpy, $v/2$ is the specific kinetic energy, and gz is the specific potential energy at the inlet (i) and outlet (e) of the control volume.

Assuming steady state $dE/dt = 0$, no heat transfer $\dot{Q} = 0$ (adiabatic process, thus $h_i = h_e$), negligible change in potential energy ($z_i \approx z_e$), and that air stops completely at the turbine blades ($v_e = 0$), we can simplify the equation to:

$$\dot{W} = \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2\right] - \sum \dot{m}_e \left[h_e\right] = \dot{m} \cdot \frac{1}{2}v^2$$

The same result as before.

2-30E

2–30E A spring whose spring constant is 200 lbf/in has an initial force of 100 lbf acting on it. Determine the work, in Btu, required to compress it another 1 in.

Given:

- $k = 200 lbf/in$
- $F_0 = 100 lbf$

Find:

- Work in Btw to compress the spring 1 more inch

$$F = kx_0 \implies x_0 = \frac{F_0}{k} = \frac{100\,lbf}{200\,\frac{lbf}{in}} = 0.5\,in$$

$$\Delta U_s = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_0^2 = \frac{1}{2}\left[200\,\frac{lbf}{in}\right][1.5\,in]^2 - \frac{1}{2}\left[200\,\frac{lbf}{in}\right][0.5\,in]^2 = 200\,lbf \cdot in$$

$$[200\,lbf \cdot in]\left[\frac{ft}{12in}\right]\left[\frac{Btu}{778.17\,ft \cdot lbf}\right] = \boxed{0.0214\,Btu}$$

Note: $1\,Btu = 778.17\,ft \cdot lbf = 1055.06\,J$

2-45E

2–45E A water pump increases the water pressure from 15 psia to 70 psia. Determine the power input required, in hp, to pump 0.8 ft^3/s of water. Does the water temperature at the inlet have any significant effect on the required flow power? *Answer:* 11.5 hp

Given:

- Input pressure: $P_i = 15$ psia
- Output pressure: $P_e = 70$ psia
- Volume flow rate $\dot{V} = 0.8 ft^3/s$

Find:

- Input power required in Hp
- Is the water temp at the inlet negligable?

Note: Psia means pounds per square inch absolute, which is the pressure relative to a perfect vacuum.

Using flow power formula

$$\boxed{\dot{W} = \dot{V}(P_e - P_i)}$$

$$\dot{W} = \left[0.8 \frac{ft^3}{s}\right] \left[70 - 15 \frac{lbf}{in^2}\right] \left[\frac{(12\, in)^2}{ft^2}\right] = 6336 \frac{lbf \cdot ft}{s}$$

$$\left[6336 \frac{lbf \cdot ft}{s}\right] \left[\frac{hp}{550 \frac{lbf \cdot ft}{s}}\right] = \boxed{11.52\, hp}$$

Note: $1\, hp = 550 \frac{lbf \cdot ft}{s} = 745.7\, W$

2-92

2–92 The inner and outer surfaces of a 0.5-cm-thick 2-m × 2-m window glass in winter are 15°C and 6°C, respectively. If the thermal conductivity of the glass is 0.78 W/m·°C, determine the amount of heat loss, in kJ, that occurs through the glass over a period of 10 h. What would your answer be if the glass were 1 cm thick?

Given:

- Thickness of window $\Delta x = 0.5$ cm
- Area of window 2 m × 2 m
- Temp inside $T_i = 15°C$
- Temp outside $T_e = 6°C$
- Thermal conductivity of glass $k = 0.78$ W/m·$°C$

Find:

- Find heat loss in kJ over 10 hrs
- Find heat loss is thickness is 1 cm

Heat conduction formula

$$\boxed{\dot{Q}_{\text{cond}} = kA\frac{\Delta T}{\Delta x}}$$

Where: $\Delta T = T_i - T_e$, A is the area of the window, k is the thermal conductivity of the material, and Δx is the thickness of the window.

$$\dot{Q} = \left[0.78\,\frac{W}{m\cdot^{\circ}C}\right]\left[4\,m^2\right]\frac{[15-6\,^{\circ}C]}{[0.005\,m]} = 5616\,W$$

$$\Longrightarrow\; Q = \dot{Q}\Delta t = [5616\,W][10\,hrs]\left[\frac{3600\,s}{1\,hr}\right] = \boxed{2.02\times 10^{8}\,J}$$

Doubling the thickness Δx to 1 cm halves the heat loss, so $Q = 1.01 \times 10^8\,J$

2-119

2–119 Consider a vertical elevator whose cabin has a total mass of 800 kg when fully loaded and 150 kg when empty. The weight of the elevator cabin is partially balanced by a 400-kg counterweight that is connected to the top of the cabin by cables that pass through a pulley located on top of the elevator well. Neglecting the weight of the cables and assuming the guide rails and the pulleys to be frictionless, determine (*a*) the power required while the fully loaded cabin is rising at a constant speed of 1.2 m/s and (*b*) the power required while the empty cabin is descending at a constant speed of 1.2 m/s. What would your answer be to (*a*) if no counterweight were used? What would your answer be to (*b*) if a friction force of 800 N has developed between the cabin and the guide rails?

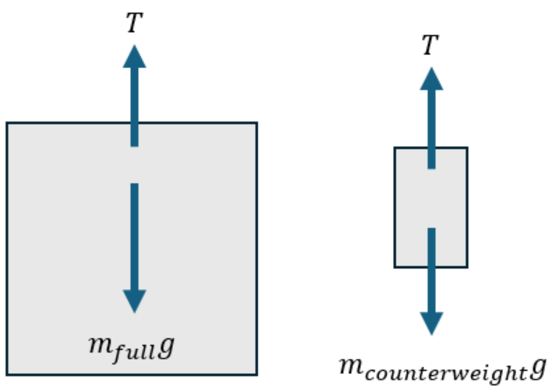
Given:

- $m_{\text{full}} = 800$ kg
- $m_{\text{empty}} = 150$ kg
- $m_{\text{counterweight}} = 400$ kg

Find:

- a) Power required to lift the full elevator up at 1.2 m/s
- b) Power required to lower the empty elevator down at 1.2 m/s
- What if no counterweight for a)
- What if there was friction of 800N for b)

(a)



$$\implies T = m_{\text{counterweight}}g$$

$$\sum F_y = m_{\text{full}}g - m_{\text{counterweight}}g = [800\,kg]\left[9.81\,\frac{m}{s^2}\right] - [400\,kg]\left[9.81\,\frac{m}{s^2}\right] = 3924\,N$$

$$\dot{W} = F \cdot v = [3924\,N][1.2\,\frac{m}{s}] = \boxed{4708.8\,W}$$

(b)

$$\sum F_y = m_{\text{empty}}g - m_{\text{counterweight}}g = [150\,kg]\left[9.81\,\frac{m}{s^2}\right] - [400\,kg]\left[9.81\,\frac{m}{s^2}\right] = -2452.5\,N$$

$$\dot{W} = F \cdot v = [-2452.5\,N][1.2\,\frac{m}{s}] = \boxed{-2943\,W}$$

No counterweight for a)

$$\implies F = 7848\,N$$

$$\dot{W} = F \cdot v = [7848\,N][1.2\,\frac{m}{s}] = \boxed{9417.6\,W}$$

Friction of 800N for b)

$$\sum F_y = m_{\text{empty}} g - m_{\text{counterweight}} g - F_f = [150\,kg]\left[9.81\,\frac{m}{s^2}\right] - [400\,kg]\left[9.81\,\frac{m}{s^2}\right] - [800\,N] = -3252.5\,N$$

$$\dot{W} = F \cdot v = [-3252.5\,N][1.2\,\frac{m}{s}] = \boxed{-3903\,W}$$

2-129

2–129 A fan is to accelerate quiescent air to a velocity of 9 m/s at a rate of 3 m^3/s. If the density of air is 1.15 kg/m^3, the minimum power that must be supplied to the fan is
(*a*) 41 W (*b*) 122 W (*c*) 140 W
(*d*) 206 W (*e*) 280 W

Given:

- $v = 9$ m/s
- Volume flow rate $\dot{V} = 3$ m^3/s
- Density $\rho = 1.15$ kg/m^3

Find:

- Min power supplied to the fan

Like problem 2-17,

$$\dot{W} = \dot{m} e_{\text{mech}} = \frac{1}{2}\rho \dot{V} v^2$$

$$\dot{W} = \frac{1}{2}[1.15\,\frac{kg}{m^3}][3\,m^3/s][9\,m/s]^2 = \boxed{139.7\,W}$$